

TASK 2 Predictive maintenance

Let us take a quick look at the data, we have a target y, column fail that indicates failure. X is an input dataset with features except the fail column. We use XGboost to predict failure. We divide the input dataset into training and test datasets. The input dataset is highly imbalanced so we use SMOTE to create synthetic data to balance the dataset.

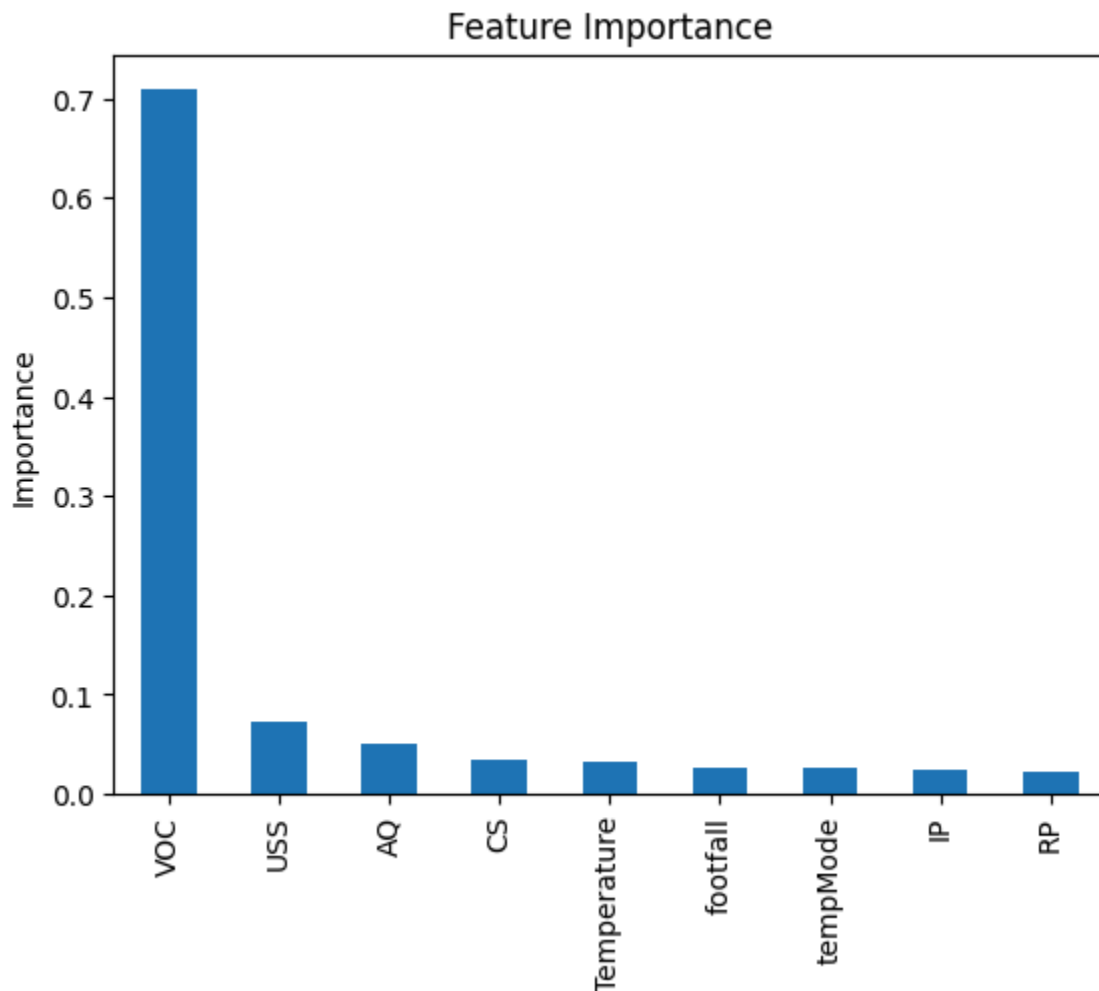
To evaluate model performance we use classification reports containing precision, recall and F1s score. We also plot confusion matrix, ROC-AUC and precision recall curves. Confusion matrix shows correct vs incorrect predictions.

ROC-AUC measures a model's ability to separate classes.

PR-AUC is useful when classes are imbalanced.

Cross-validation F1 shows stability across folds.

Feature importance: VOC shows highest correlation with predicted output



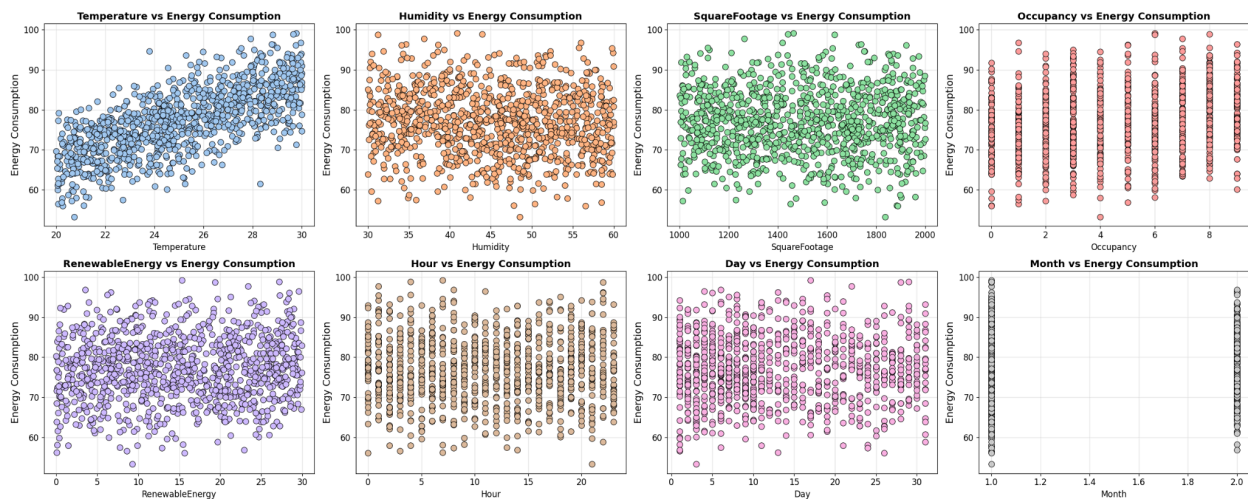
TASK 3 Energy forecasting:

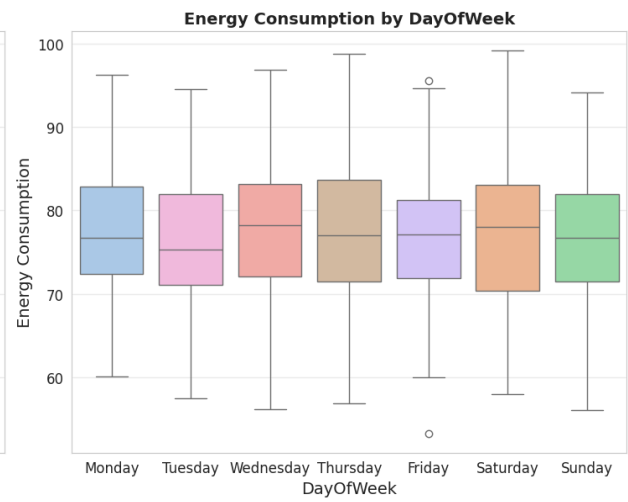
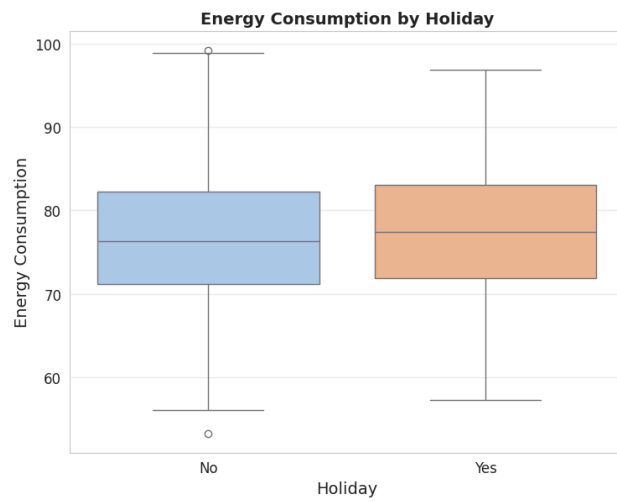
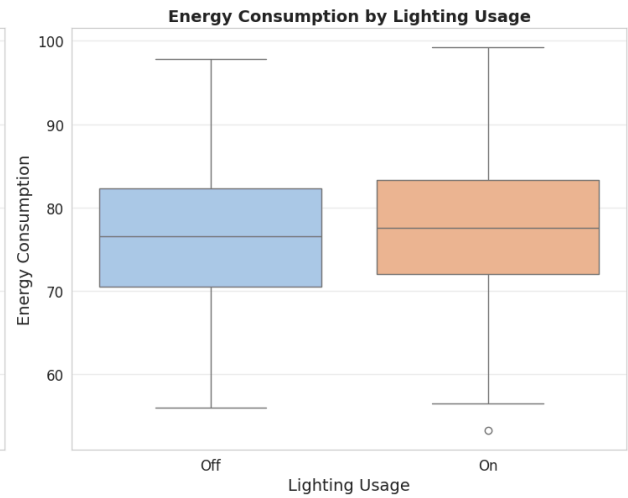
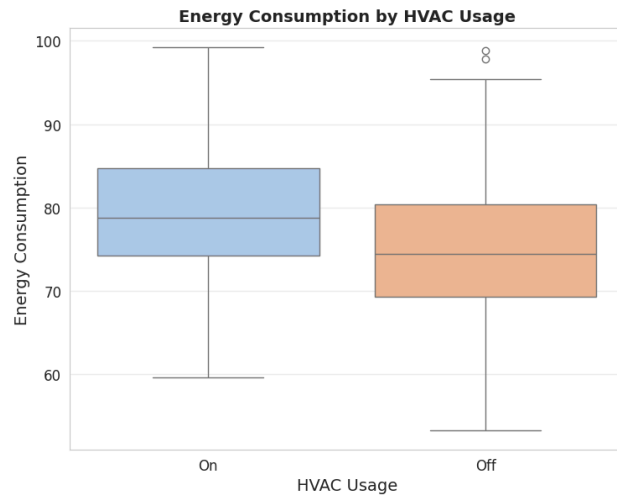
Let us take a look at the input dataset

```
Numerical features: Index(['Temperature', 'Humidity',  
    'SquareFootage', 'Occupancy', 'RenewableEnergy', 'EnergyConsumption',  
    'Hour', 'Day', 'Month'], dtype='object')
```

```
Categorical features: Index(['HVACUsage', 'LightingUsage',  
    'DayOfWeek', 'Holiday'], dtype='object')
```

We plot EnergyConsumption against each of the categorical and numeric features



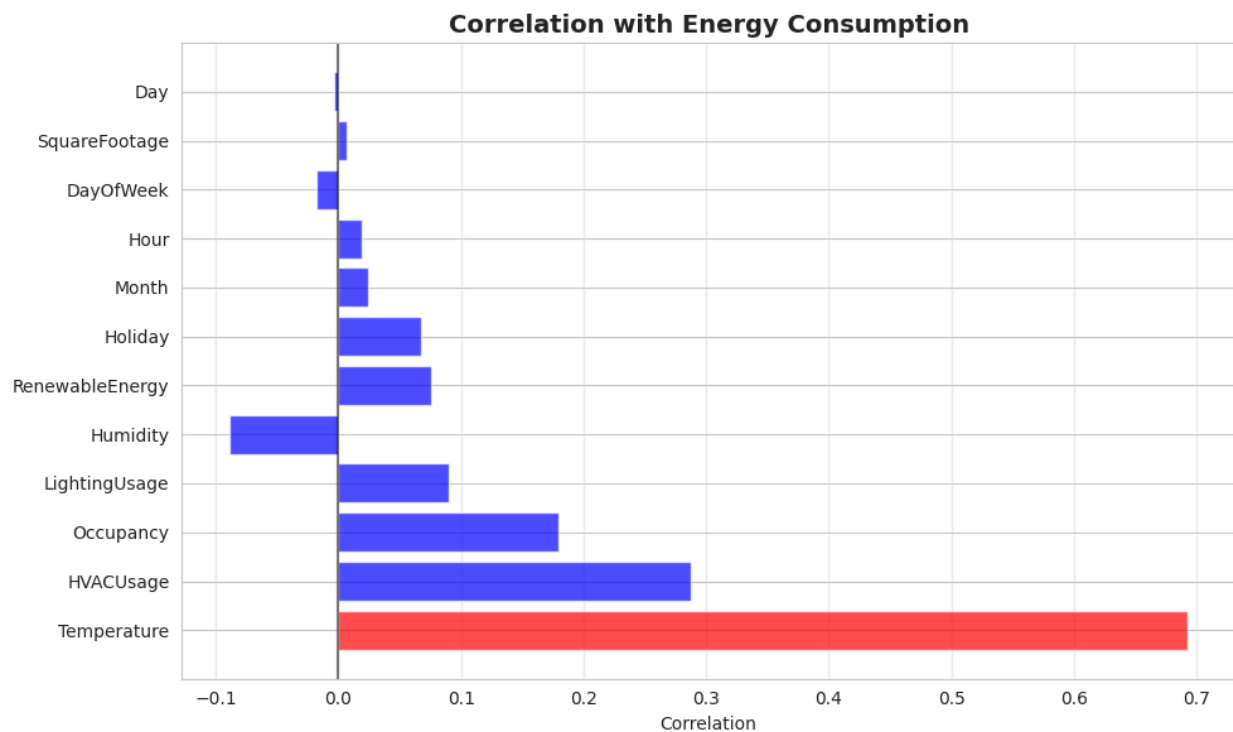


Data Processing Finding Outliers:

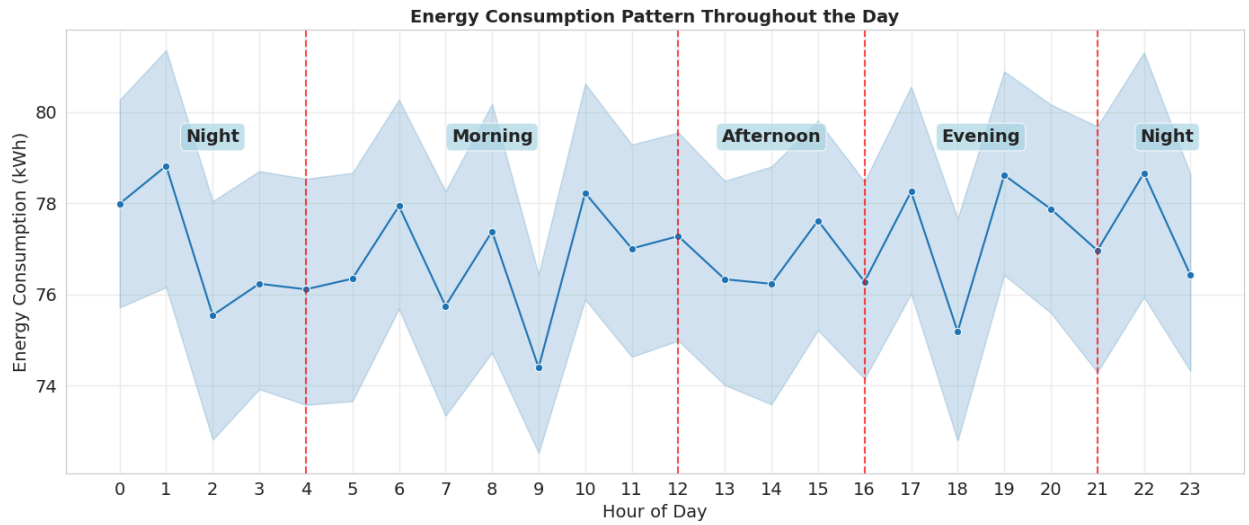
The input dataset contains several outliers. Remove outliers in categorical variables based on IQR range. Remove outliers in numeric features based on absolute z score greater than 3.

Feature Engineering

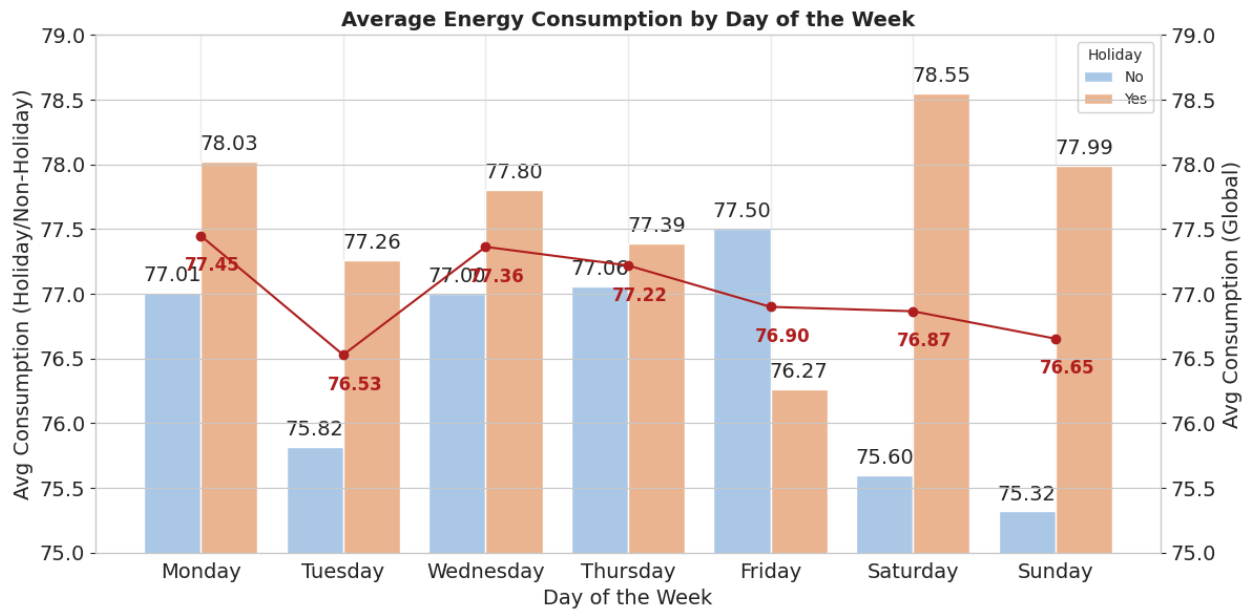
1. Convert categorical labels into numerical values.
For ordinal variables use label encoder on = 1 off = 0
For categorical variables use one hot encoding eg day of the week 1 to 7
2. Calculate correlation with energy consumption after removing outliers



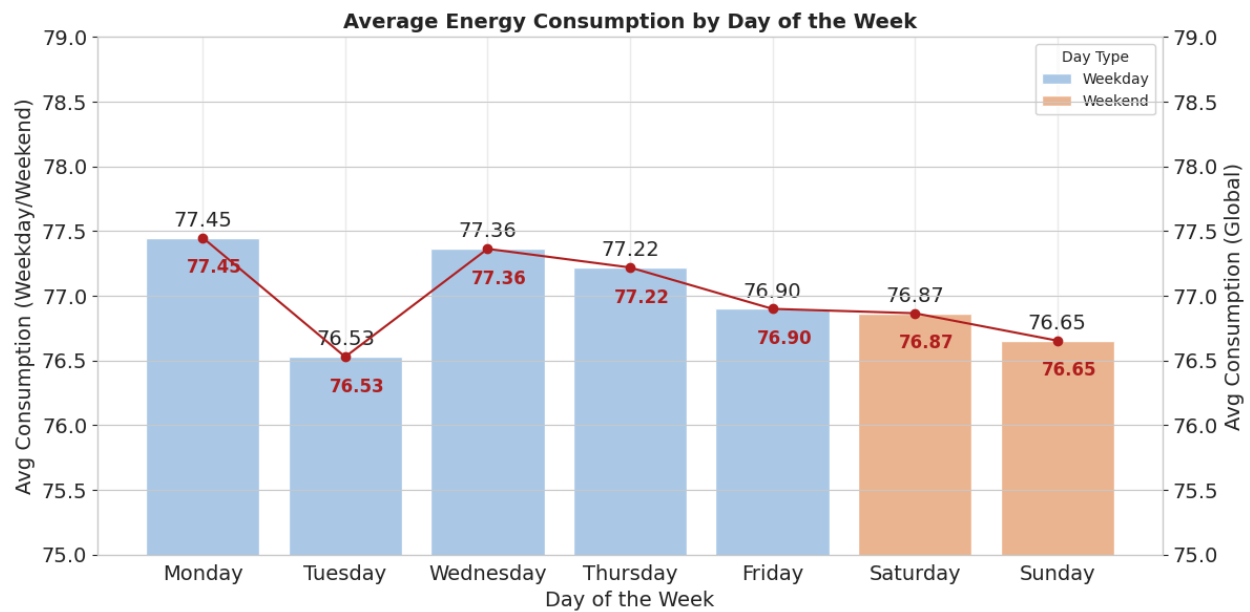
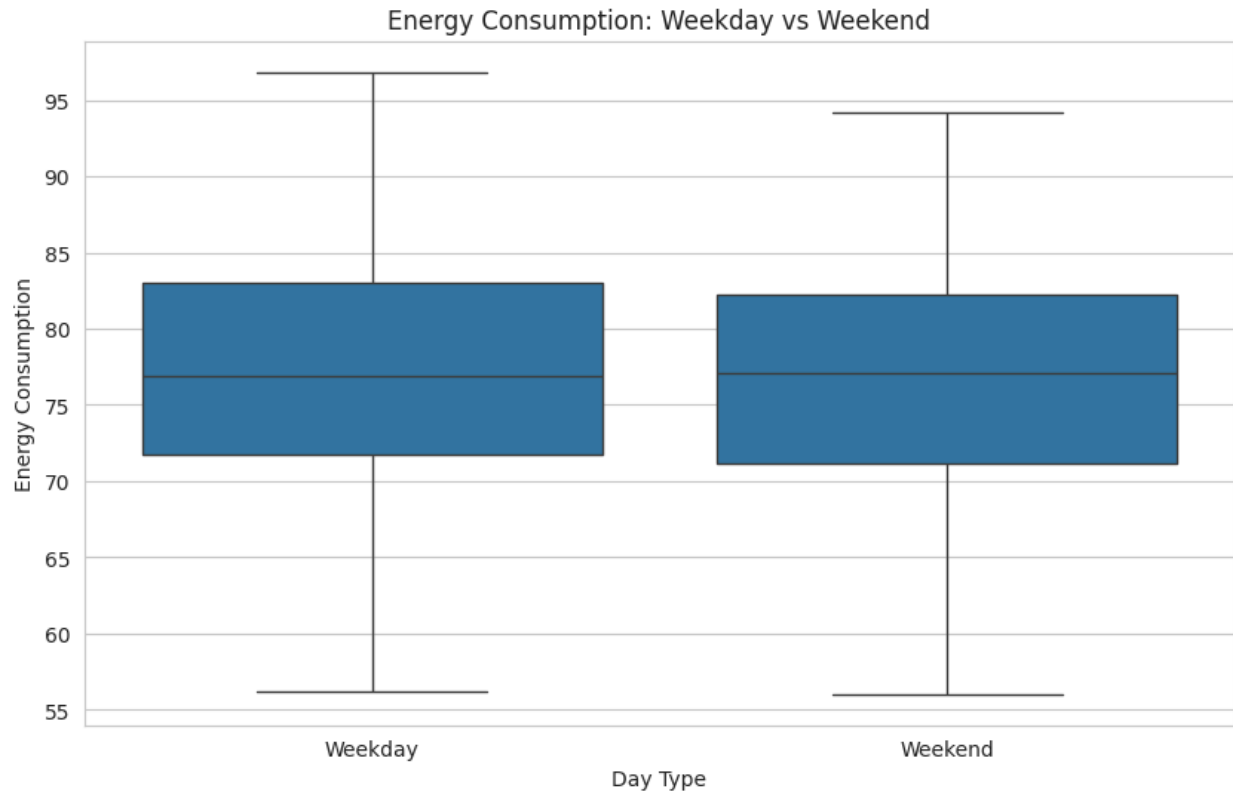
3. **New Feature:** Convert hour column into time of day: Morning, afternoon, evening and night



4. Visualize effect of holidays



5. Visualize effect of weekdays vs weekend



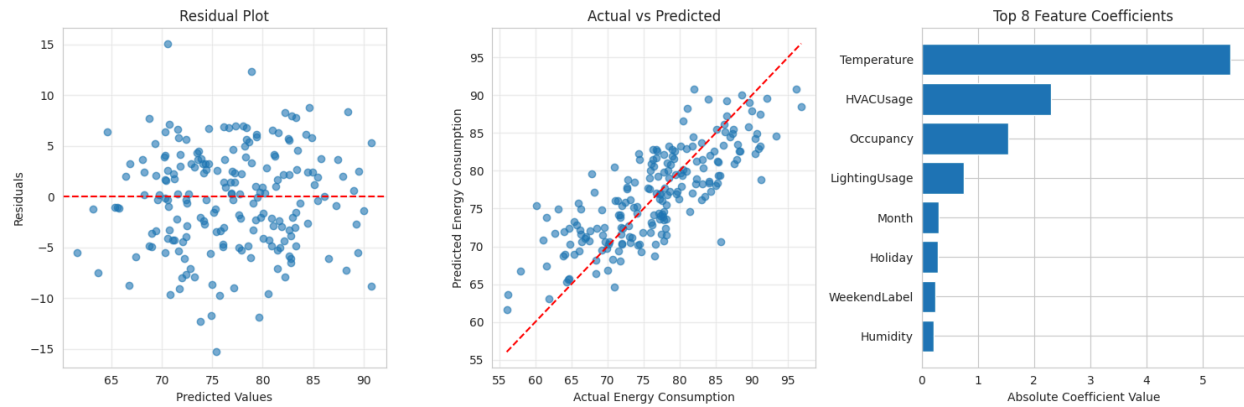
6. **New Feature:** Add new feature weekendlabel = true if day is saturday or sunday

```
features = ['Temperature', 'Humidity', 'SquareFootage',
'Occupancy', 'HVACUsage', 'LightingUsage', \
            'DayOfWeek', 'Holiday', 'Hour', 'Day', 'Month', 'WeekendLabel',
'TimePeriodLabel']
```

In total, we added two new features **'WeekendLabel'** , **'TimePeriodLabel'**

7. Compare models RandomForest, XGBoost, linear regression and ridge regression based on RMSE values. Linear regression provides the best fit.

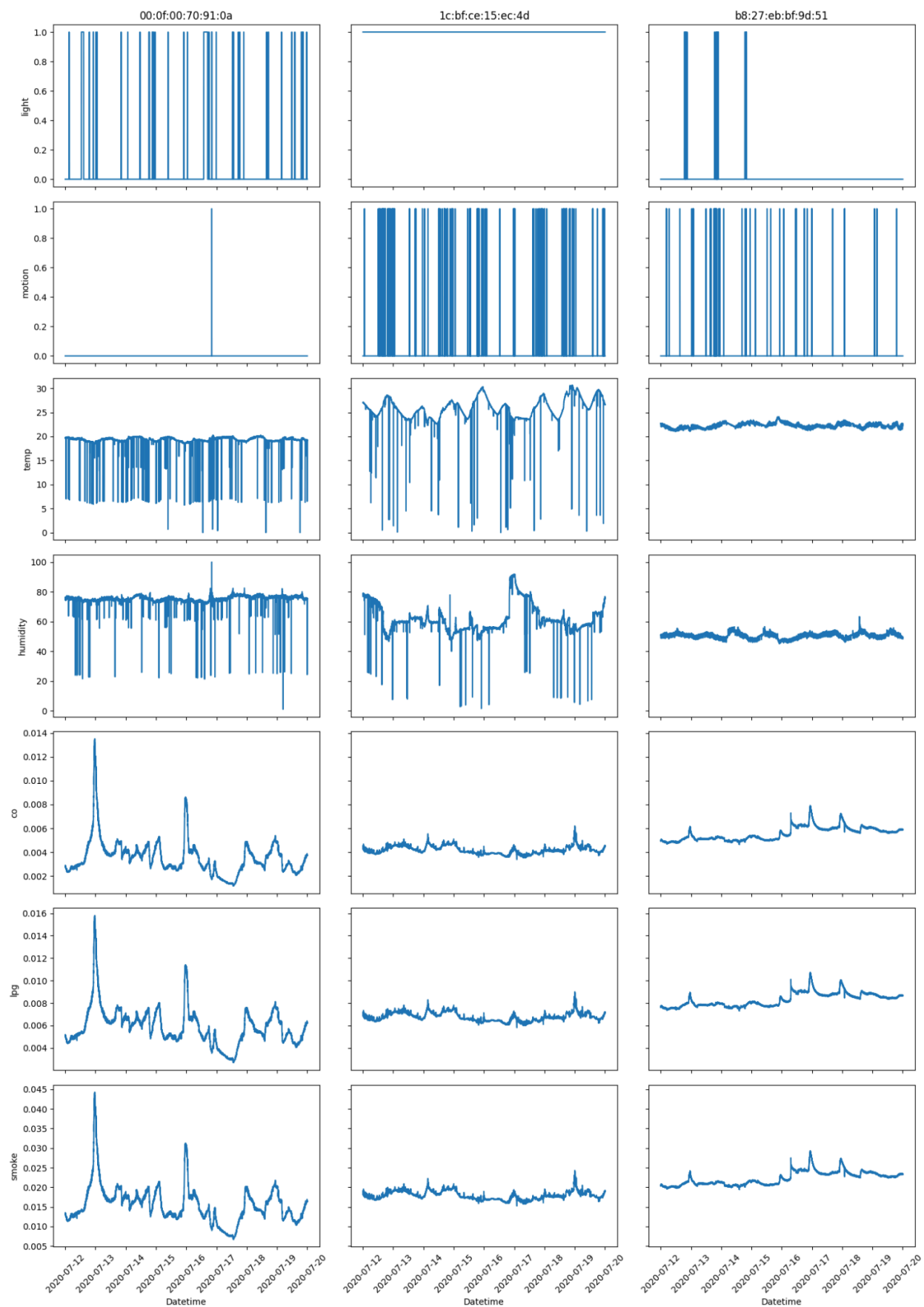
Feature importance



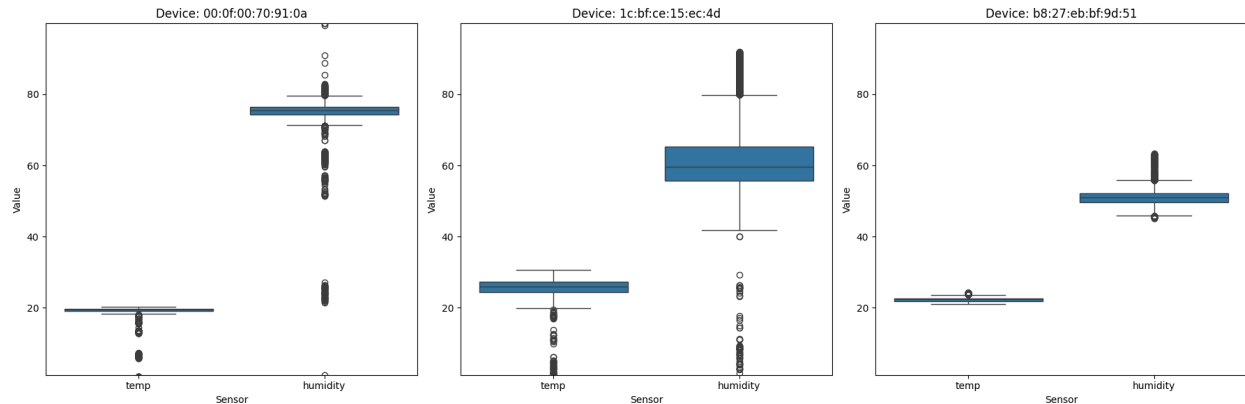
TASK 1: EDA:

Let us take a look at input dataset. The sensors data include:

```
sensors = ['light', 'motion', 'temp', 'humidity', 'co', 'lpg', 'smoke']
```

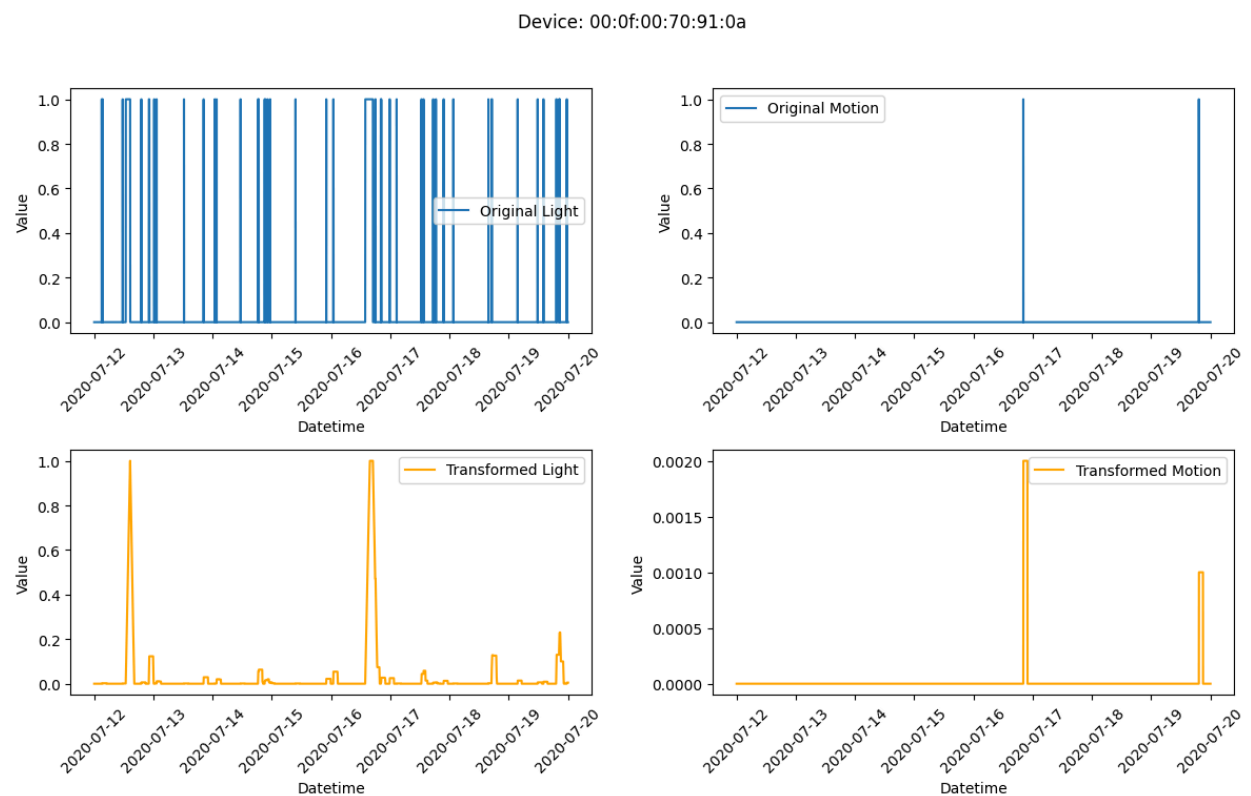


We remove smoke and lpg sensors, they are measured by CO sensors.
There are three devices in total. For each we plot temp and humidity. Device0, device1, device2.
Device 1 shows highest variability while 0 and 2 are placed in stable/well controlled temp and humidity environments.



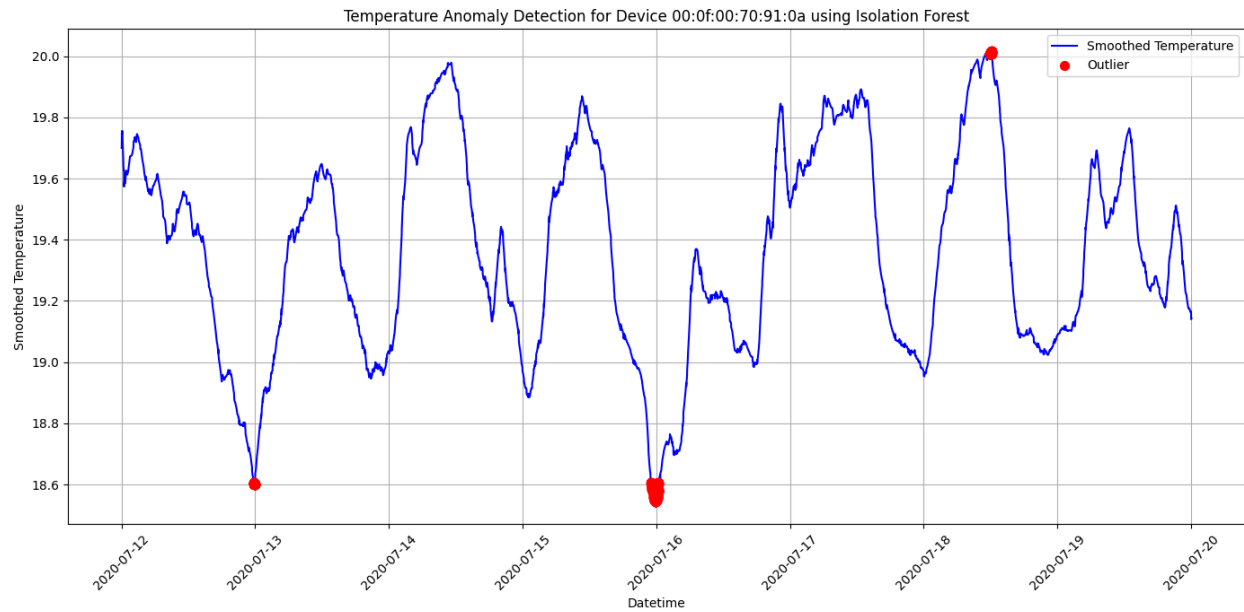
Binary data processing:

The light and motion readouts are categorical columns, implementing rolling average to smooth out epochs for later use in anomaly detection using isolation forest.



Task 4: Anomaly detection Isolation Forest

device 00 temp anomaly



Device 00 Motion anomaly

